IQBAL_INTISEIER_382_01_CPSPROJECT

Data Project

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#CPS_DATA_EXCERCISE

```
### RUN THIS FIRST TO DOWNLOAD NEEDED PACKAGES (Press Yes to all)

options(repos = c(CRAN = "https://cloud.r-project.org/"))

CPSDATA <- read.csv("CPSDATA.csv")

install.packages("lmtest")
```

```
Installing package into 'C:/Users/otrim/AppData/Local/R/win-library/4.4'
(as 'lib' is unspecified)

package 'lmtest' successfully unpacked and MD5 sums checked

Warning: cannot remove prior installation of package 'lmtest'

Warning in file.copy(savedcopy, lib, recursive = TRUE): problem copying
C:\Users\otrim\AppData\Local\R\win-library\4.4\00LOCK\lmtest\libs\x64\lmtest.dll
to C:\Users\otrim\AppData\Local\R\win-library\4.4\lmtest\libs\x64\lmtest.dll:
Permission denied

Warning: restored 'lmtest'


The downloaded binary packages are in
    C:\Users\otrim\AppData\Local\Temp\RtmpspFnXF\downloaded_packages
```

```
install.packages("sandwich")
```

```
Installing package into 'C:/Users/otrim/AppData/Local/R/win-library/4.4'
(as 'lib' is unspecified)

package 'sandwich' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
    C:\Users\otrim\AppData\Local\Temp\RtmpspFnXF\downloaded_packages
```

```
install.packages("rstudioapi", repos = "https://cloud.r-project.org/")
```

```
Installing package into 'C:/Users/otrim/AppData/Local/R/win-library/4.4'
(as 'lib' is unspecified)

package 'rstudioapi' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
    C:\Users\otrim\AppData\Local\Temp\RtmpspFnXF\downloaded_packages
```

```
install.packages("https://cran.rstudio.com/bin/windows/contrib/4.4/psych_2.4.6.26.zip")
```

```
Installing package into 'C:/Users/otrim/AppData/Local/R/win-library/4.4'
(as 'lib' is unspecified)

inferring 'repos = NULL' from 'pkgs'

package 'psych' successfully unpacked and MD5 sums checked
```

```
install.packages("zoo", repos = "https://cloud.r-project.org/")
```

```
Installing package into 'C:/Users/otrim/AppData/Local/R/win-library/4.4'
(as 'lib' is unspecified)

package 'zoo' successfully unpacked and MD5 sums checked

Warning: cannot remove prior installation of package 'zoo'

Warning in file.copy(savedcopy, lib, recursive = TRUE): problem copying
C:\Users\otrim\AppData\Local\R\win-library\4.4\00LOCK\zoo\libs\x64\zoo.dll to
C:\Users\otrim\AppData\Local\R\win-library\4.4\zoo\libs\x64\zoo.dll: Permission
denied

Warning: restored 'zoo'


The downloaded binary packages are in
    C:\Users\otrim\AppData\Local\Temp\RtmpspFnXF\downloaded_packages
```

```
library(zoo) #may get error because of same function name in another package
```

```
Attaching package: 'zoo'

The following objects are masked from 'package:base':

    as.Date, as.Date.numeric
```

```
library(lmtest)
library(sandwich)
library(psych)
```

```r
library(rstudioapi)

# Using these to check if data is right and column number are there.

# View the structure of the dataset
str(CPSDATA)
```

```
'data.frame':   146133 obs. of  20 variables:
 $ YEAR       : int  2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
 $ SERIAL     : int  1 1 2 2 10 11 11 12 12 13 ...
 $ MONTH      : int  3 3 3 3 3 3 3 3 3 3 ...
 $ CPSID      : num  2.02e+13 2.02e+13 2.02e+13 2.02e+13 2.02e+13 ...
 $ ASECFLAG   : int  1 1 1 1 1 1 1 1 1 1 ...
 $ ASECWTH    : num  1580 1580 1790 1790 1308 ...
 $ REGION     : int  11 11 11 11 11 11 11 11 11 11 ...
 $ STATEFIP   : int  23 23 23 23 23 23 23 23 23 23 ...
 $ STATECENSUS: int  11 11 11 11 11 11 11 11 11 11 ...
 $ COUNTY     : int  0 0 0 0 0 0 0 0 0 0 ...
 $ PERNUM     : int  1 2 1 2 1 1 2 1 2 1 ...
 $ CPSIDV     : num  2.02e+14 2.02e+14 2.02e+14 2.02e+14 2.02e+14 ...
 $ CPSIDP     : num  2.02e+13 2.02e+13 2.02e+13 2.02e+13 2.02e+13 ...
 $ ASECWT     : num  1580 1580 1790 1790 1308 ...
 $ AGE        : int  66 68 52 51 78 65 68 74 74 76 ...
 $ SEX        : int  2 2 2 1 2 1 2 2 1 1 ...
 $ RACE       : int  100 100 100 100 100 100 100 100 100 100 ...
 $ MARST      : int  5 5 1 1 5 1 1 1 1 1 ...
 $ EDUC99     : int  10 10 10 11 10 11 13 5 5 16 ...
 $ INCWAGE    : int  0 0 0 42000 0 55000 52000 0 0 0 ...
```

```r
# Look at the first few rows
head(CPSDATA)
```

```
  YEAR SERIAL MONTH       CPSID ASECFLAG ASECWTH REGION STATEFIP STATECENSUS
1 2023      1     3 2.02201e+13        1 1580.07     11       23          11
2 2023      1     3 2.02201e+13        1 1580.07     11       23          11
3 2023      2     3 2.02301e+13        1 1789.60     11       23          11
4 2023      2     3 2.02301e+13        1 1789.60     11       23          11
5 2023     10     3 2.02202e+13        1 1307.92     11       23          11
6 2023     11     3 2.02212e+13        1 1307.41     11       23          11
  COUNTY PERNUM      CPSIDV      CPSIDP  ASECWT AGE SEX RACE MARST EDUC99
1      0      1 2.02201e+14 2.02201e+13 1580.07  66   2  100     5     10
2      0      2 2.02201e+14 2.02201e+13 1580.07  68   2  100     5     10
3      0      1 2.02301e+14 2.02301e+13 1789.60  52   2  100     1     10
4      0      2 2.02301e+14 2.02301e+13 1789.60  51   1  100     1     11
5      0      1 2.02202e+14 2.02202e+13 1307.92  78   2  100     5     10
6      0      1 2.02212e+14 2.02212e+13 1307.41  65   1  100     1     11
  INCWAGE
1       0
2       0
3       0
```

```
4   42000
5       0
6   55000
```

```
# Summary statistics
summary(CPSDATA)
```

```
     YEAR          SERIAL          MONTH       CPSID              ASECFLAG
 Min.   :2023   Min.   :    1   Min.   :3   Min.   :0.000e+00   Min.   :1
 1st Qu.:2023   1st Qu.:23069   1st Qu.:3   1st Qu.:0.000e+00   1st Qu.:1
 Median :2023   Median :46014   Median :3   Median :2.022e+13   Median :1
 Mean   :2023   Mean   :45803   Mean   :3   Mean   :1.333e+13   Mean   :1
 3rd Qu.:2023   3rd Qu.:68796   3rd Qu.:3   3rd Qu.:2.023e+13   3rd Qu.:1
 Max.   :2023   Max.   :88978   Max.   :3   Max.   :2.023e+13   Max.   :1
    ASECWTH            REGION         STATEFIP     STATECENSUS
 Min.   :  117.2   Min.   :11.00   Min.   : 1.0   Min.   :11.00
 1st Qu.: 1132.2   1st Qu.:21.00   1st Qu.:12.0   1st Qu.:35.00
 Median : 2114.1   Median :31.00   Median :28.0   Median :59.00
 Mean   : 2181.4   Mean   :29.56   Mean   :27.7   Mean   :58.83
 3rd Qu.: 2873.7   3rd Qu.:41.00   3rd Qu.:42.0   3rd Qu.:83.00
 Max.   :10925.2   Max.   :42.00   Max.   :56.0   Max.   :95.00
     COUNTY          PERNUM           CPSIDV              CPSIDP
 Min.   :    0   Min.   : 1.000   Min.   :0.000e+00   Min.   :0.000e+00
 1st Qu.:    0   1st Qu.: 1.000   1st Qu.:0.000e+00   1st Qu.:0.000e+00
 Median :    0   Median : 2.000   Median :2.022e+14   Median :2.022e+13
 Mean   : 9439   Mean   : 2.215   Mean   :1.333e+14   Mean   :1.333e+13
 3rd Qu.:13135   3rd Qu.: 3.000   3rd Qu.:2.023e+14   3rd Qu.:2.023e+13
 Max.   :55139   Max.   :16.000   Max.   :2.023e+14   Max.   :2.023e+13
     ASECWT             AGE             SEX             RACE
 Min.   :  117.2   Min.   : 0.00   Min.   :1.000   Min.   :100.0
 1st Qu.: 1148.2   1st Qu.:18.00   1st Qu.:1.000   1st Qu.:100.0
 Median : 2148.2   Median :38.00   Median :2.000   Median :100.0
 Mean   : 2262.5   Mean   :38.73   Mean   :1.512   Mean   :176.4
 3rd Qu.: 3015.1   3rd Qu.:58.00   3rd Qu.:2.000   3rd Qu.:100.0
 Max.   :14588.1   Max.   :85.00   Max.   :2.000   Max.   :830.0
     MARST           EDUC99          INCWAGE
 Min.   :1.000   Min.   : 0.00   Min.   :       0
 1st Qu.:1.000   1st Qu.: 6.00   1st Qu.:       0
 Median :4.000   Median :10.00   Median :   36000
 Mean   :3.711   Mean   : 9.41   Mean   :20206913
 3rd Qu.:6.000   3rd Qu.:15.00   3rd Qu.:  125000
 Max.   :6.000   Max.   :18.00   Max.   :99999999
```

```
### RUN THIS AFTER DOWNLOADING NEEDED PACKAGES ABOVE

library(zoo) #may get error because of same function name in another package
library(lmtest)
library(sandwich)
library(psych)
library(rstudioapi)
```

```
CPSDATA <- read.csv("CPSDATA.csv")


# (7) Create a FEMALE dummy  [0 <- Male, 1 <- Female]
CPSDATA$FEMALE <- ifelse(
  CPSDATA$SEX == 2, 1, 0
  )
table(CPSDATA$FEMALE, useNA = "ifany")
```

```
    0     1 
71299 74834 
```

```
# (8) Create NONWHITE dummy [0 <- White, 1 <- Non White]
CPSDATA$NONWHITE <- ifelse(
    CPSDATA$RACE == 999, NA, ifelse(
      CPSDATA$RACE != 100, 1, 0
  )
)
table(CPSDATA$NONWHITE, useNA = "ifany")
```

```
     0      1 
111621  34512 
```

```
# (9) Create MARRIED dummy [0 <- Not Married, 1 <- Married]
CPSDATA$MARRIED <- ifelse(
  CPSDATA$MARST == 9, NA, ifelse(
    CPSDATA$MARST == 1, 1, 0
  )
)
table(CPSDATA$MARRIED, useNA = "ifany")
```

```
    0     1 
87497 58636 
```

```
# (10) Create HIGHSCHOOL dummy [1 <- High school graduate or higher]
#                              [0 <- Less than high school]
CPSDATA$HIGHSCHOOL <- ifelse(
  CPSDATA$EDUC99 == 0, NA, ifelse(
    CPSDATA$EDUC99 >= 10, 1, 0
  )
)
table(CPSDATA$HIGHSCHOOL, useNA = "ifany")
```

```
    0     1  <NA> 
```

```
17573 99077 29483
```

```r
# (11) Handle missing values in INCWAGE then Verify
CPSDATA$INCWAGE <- ifelse(
  CPSDATA$INCWAGE == 99999998 | CPSDATA$INCWAGE == 99999999, NA, CPSDATA$INCWAGE
)
summary(CPSDATA$INCWAGE)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
      0       0   17600   39407   55000 1549999   29483 
```

```r
# (12) Region variable into 4 groups then Verify
CPSDATA$REGION <- ifelse(
  CPSDATA$REGION == 11 | CPSDATA$REGION == 12, 1,
  ifelse(CPSDATA$REGION == 21 | CPSDATA$REGION == 22, 2,
  ifelse(CPSDATA$REGION == 31 | CPSDATA$REGION == 32 | CPSDATA$REGION == 33, 3,
  ifelse(CPSDATA$REGION == 41 | CPSDATA$REGION == 42, 4, NA)))
)
table(CPSDATA$REGION, useNA = "ifany")
```

```
    1     2     3     4 
22386 27411 54526 41810 
```

```r
# (13) Drop unnecessary columns: SEX, RACE, MARST, and EDUC99. Then Verify.
CPSDATA <- CPSDATA[, !(names(CPSDATA) %in% c("SEX", "RACE", "MARST", "EDUC99"))]

names(CPSDATA)
```

```
 [1] "YEAR"        "SERIAL"      "MONTH"       "CPSID"       "ASECFLAG"   
 [6] "ASECWTH"     "REGION"      "STATEFIP"    "STATECENSUS" "COUNTY"     
[11] "PERNUM"      "CPSIDV"      "CPSIDP"      "ASECWT"      "AGE"        
[16] "INCWAGE"     "FEMALE"      "NONWHITE"    "MARRIED"     "HIGHSCHOOL"
```

```r
# (14) Summary statistics for selected variables
summary_data <- summary(
  CPSDATA[, c("INCWAGE", "FEMALE", "NONWHITE", "MARRIED", "HIGHSCHOOL", "REGION")]
  )

print(summary_data)
```

```
   INCWAGE            FEMALE          NONWHITE         MARRIED      
 Min.   :      0   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000  
 1st Qu.:      0   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000  
 Median :  17600   Median :1.0000   Median :0.0000   Median :0.0000  
 Mean   :  39407   Mean   :0.5121   Mean   :0.2362   Mean   :0.4013  
 3rd Qu.:  55000   3rd Qu.:1.0000   3rd Qu.:0.0000   3rd Qu.:1.0000  
 Max.   :1549999   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000  
```

```
 NA's   :29483
  HIGHSCHOOL        REGION
 Min.   :0.000   Min.   :1.000
 1st Qu.:1.000   1st Qu.:2.000
 Median :1.000   Median :3.000
 Mean   :0.849   Mean   :2.792
 3rd Qu.:1.000   3rd Qu.:4.000
 Max.   :1.000   Max.   :4.000
 NA's   :29483
```

```
# (15) Calculate the fraction of zero values in INCWAGE
zero_incwage_fraction <- mean(CPSDATA$INCWAGE == 0, na.rm = TRUE)

# Print the fraction of zero values then Verify
print(paste(
  "Fraction of zero values that are Non-NA in INCWAGE:", round(zero_incwage_fraction, 4), "out of
  )
```

```
[1] "Fraction of zero values that are Non-NA in INCWAGE: 0.4072 out of 1"
```

```
CPSDATA$INCWAGE[CPSDATA$INCWAGE == 0] <- 0.01

sum(CPSDATA$INCWAGE == 0, na.rm = TRUE) # This should return 0
```

```
[1] 0
```

```
any(CPSDATA$INCWAGE == 0, na.rm = TRUE) # This should return FALSE
```

```
[1] FALSE
```

```
head(sort(CPSDATA$INCWAGE), 10)         # This should return 10 "0.01's"
```

```
 [1] 0.01 0.01 0.01 0.01 0.01 0.01 0.01 0.01 0.01 0.01
```

```
min(CPSDATA$INCWAGE, na.rm = TRUE)      # This should return 0.01
```

```
[1] 0.01
```

```
summary(CPSDATA$INCWAGE)      # This should return 0 in minimum because its rounded
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
      0       0   17600   39407   55000 1549999   29483
```

```
# (16) Model 1
model1 <- lm(log(INCWAGE) ~ FEMALE + NONWHITE + MARRIED + HIGHSCHOOL, data = CPSDATA)
```

```r
summary(model1)
```

```
Call:
lm(formula = log(INCWAGE) ~ FEMALE + NONWHITE + MARRIED + HIGHSCHOOL,
    data = CPSDATA)

Residuals:
    Min      1Q  Median      3Q     Max 
-11.010  -8.561   4.510   5.948  14.723 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  0.88668    0.06131   14.46  < 2e-16 ***
FEMALE      -1.54394    0.04269  -36.17  < 2e-16 ***
NONWHITE    -0.15536    0.05044   -3.08  0.00207 ** 
MARRIED      0.90518    0.04377   20.68  < 2e-16 ***
HIGHSCHOOL   4.61319    0.06085   75.81  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.274 on 116645 degrees of freedom
  (29483 observations deleted due to missingness)
Multiple R-squared:  0.06695,	Adjusted R-squared:  0.06692 
F-statistic:  2092 on 4 and 116645 DF,  p-value: < 2.2e-16
```

```r
robust_se <- sqrt(diag(vcovHC(model1, type = "HC1")))

# Table 1
coefs <- coef(model1)
robust_results1 <- data.frame(
  Estimate = coefs,
  `Robust SE` = robust_se,
  `t value` = coefs / robust_se,
  `Pr(>|t|)` = 2 * pt(-abs(coefs / robust_se), df = df.residual(model1))
)
# Robust.SE Listed in Column 2
summary(robust_results1)
```

```
   Estimate         Robust.SE          t.value           Pr...t..      
 Min.   :-1.5439   Min.   :0.04262   Min.   :-36.227   Min.   :0.0000000  
 1st Qu.:-0.1554   1st Qu.:0.04419   1st Qu.: -3.086   1st Qu.:0.0000000  
 Median : 0.8867   Median :0.05034   Median : 15.407   Median :0.0000000  
 Mean   : 0.9412   Mean   :0.05041   Mean   : 15.396   Mean   :0.0004056  
 3rd Qu.: 0.9052   3rd Qu.:0.05738   3rd Qu.: 20.486   3rd Qu.:0.0000000  
 Max.   : 4.6132   Max.   :0.05755   Max.   : 80.402   Max.   :0.0020281  
```

```r
# (17) Model 2
model2 <- lm(log(INCWAGE) ~
               FEMALE + NONWHITE + MARRIED + HIGHSCHOOL + FEMALE:NONWHITE,
             data = CPSDATA)

summary(model2)
```

```
Call:
lm(formula = log(INCWAGE) ~ FEMALE + NONWHITE + MARRIED + HIGHSCHOOL +
    FEMALE:NONWHITE, data = CPSDATA)

Residuals:
    Min      1Q  Median      3Q     Max
-11.113  -8.464   4.494   5.960  14.667

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      0.98693    0.06244  15.807   <2e-16 ***
FEMALE          -1.74310    0.04878 -35.733   <2e-16 ***
NONWHITE        -0.60377    0.07331  -8.236   <2e-16 ***
MARRIED          0.90576    0.04375  20.702   <2e-16 ***
HIGHSCHOOL       4.61541    0.06083  75.869   <2e-16 ***
FEMALE:NONWHITE  0.84704    0.10051   8.427   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.272 on 116644 degrees of freedom
  (29483 observations deleted due to missingness)
Multiple R-squared:  0.06752,   Adjusted R-squared:  0.06748
F-statistic:  1689 on 5 and 116644 DF,  p-value: < 2.2e-16
```

```r
robust_se2 <- sqrt(diag(vcovHC(model2, type = "HC1")))

# Table 2
coefs2 <- coef(model2)
robust_results2 <- data.frame(
  Estimate = coefs2,
  `Robust SE` = robust_se2,
  `t value` = coefs2 / robust_se2,
  `Pr(>|t|)` = 2 * pt(-abs(coefs2 / robust_se2), df = df.residual(model2))
)
# Robust.SE Listed in Column 2
summary(robust_results2)
```

```
   Estimate         Robust.SE          t.value           Pr...t..
 Min.   :-1.7431   Min.   :0.04418   Min.   :-35.756   Min.   :0.000e+00
 1st Qu.:-0.2411   1st Qu.:0.05090   1st Qu.: -4.118   1st Qu.:0.000e+00
 Median : 0.8764   Median :0.05805   Median : 12.622   Median :0.000e+00
```

```
 Mean   : 0.8347   Mean   :0.06366   Mean   : 13.696   Mean   :2.130e-17
 3rd Qu.: 0.9666   3rd Qu.:0.06920   3rd Qu.: 19.576   3rd Qu.:2.201e-17
 Max.   : 4.6154   Max.   :0.10024   Max.   : 80.490   Max.   :9.846e-17
```

```
# (18) Model 3 with regional dummies (R2, R3, R4)

CPSDATA$REGION <- factor(
  CPSDATA$REGION, levels = c(1, 2, 3, 4), labels = c("R1", "R2", "R3", "R4")
  )

# Check to see if Region is a factor and outputs R1, R2, R3, R4
class(CPSDATA$REGION)
```

```
[1] "factor"
```

```
is.factor(CPSDATA$REGION)
```

```
[1] TRUE
```

```
levels(CPSDATA$REGION)
```

```
[1] "R1" "R2" "R3" "R4"
```

```
table(CPSDATA$REGION, useNA = "ifany")
```

```
   R1    R2    R3    R4 
22386 27411 54526 41810 
```

```
model3 <- lm(log(INCWAGE) ~
        FEMALE + NONWHITE + MARRIED + HIGHSCHOOL + FEMALE:NONWHITE + REGION,
        data = CPSDATA)

# Summary of the model with standard errors
summary(model3)
```

```
Call:
lm(formula = log(INCWAGE) ~ FEMALE + NONWHITE + MARRIED + HIGHSCHOOL + 
    FEMALE:NONWHITE + REGION, data = CPSDATA)

Residuals:
    Min      1Q  Median      3Q     Max 
-11.460  -8.600   4.460   5.959  14.924 

Coefficients:
                Estimate Std. Error t value Pr(>|t|)    
(Intercept)      1.13657    0.08018  14.176  < 2e-16 ***
```

```
FEMALE          -1.74088    0.04876 -35.704  < 2e-16 ***
NONWHITE        -0.54897    0.07347  -7.472 7.93e-14 ***
MARRIED          0.90356    0.04374  20.657  < 2e-16 ***
HIGHSCHOOL       4.59860    0.06084  75.589  < 2e-16 ***
REGIONR2         0.21630    0.07356   2.940  0.00328 **
REGIONR3        -0.40853    0.06426  -6.357 2.06e-10 ***
REGIONR4        -0.12229    0.06701  -1.825  0.06801 .
FEMALE:NONWHITE  0.85056    0.10047   8.466  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.269 on 116641 degrees of freedom
  (29483 observations deleted due to missingness)
Multiple R-squared:  0.06844,   Adjusted R-squared:  0.06838
F-statistic:  1071 on 8 and 116641 DF,  p-value: < 2.2e-16
```

```r
# Calculate robust standard errors
robust_se3 <- sqrt(diag(vcovHC(model3, type = "HC1")))

# Table 3
coefs3 <- coef(model3)
robust_results3 <- data.frame(
  Estimate = coefs3,
  `Robust SE` = robust_se3,
  `t value` = coefs3 / robust_se3,
  `Pr(>|t|)` = 2 * pt(-abs(coefs3 / robust_se3), df = df.residual(model3))
)

# Robust.SE Listed in Column 2
summary(robust_results3)
```

```
   Estimate         Robust.SE          t.value           Pr...t..
 Min.   :-1.7409   Min.   :0.04417   Min.   :-35.724   Min.   :0.000000
 1st Qu.:-0.4085   1st Qu.:0.05734   1st Qu.: -6.343   1st Qu.:0.000000
 Median : 0.2163   Median :0.06719   Median :  2.955   Median :0.000000
 Mean   : 0.5428   Mean   :0.06728   Mean   :  8.370   Mean   :0.007988
 3rd Qu.: 0.9036   3rd Qu.:0.07320   3rd Qu.: 14.663   3rd Qu.:0.000000
 Max.   : 4.5986   Max.   :0.10017   Max.   : 80.197   Max.   :0.068762
```

```r
# (19)
# The Region Dummies are there take account of the Region a person is. Pretty much
# what ever the region is, that region would get counted. All the other regions
# will be turned to 0 in the regression. For example, if the person is in region 2.
# R3 and R4 would be 0 and R2 would be 1 * (whatever the estimate is, 0.21630), to
# be accounted for. The reason R1 is not seen is because its already accounted for
# in the original regression. So if the person is in Region 1, (R2, R3, R4) would
# be 0, and R1 would already be accounted for in the Regression.

# (20) Table for all Regressions (works)
modelsummary::modelsummary(
```

```
list(model1, model2, model3),
fmt = 3,
vcov = "HC1",
gof_omit = "AIC|BIC|Log|RMSE|Std",
title = "Comparison of all Models"
)
```

	(1)	(2)	(3)
(Intercept)	0.887	0.987	1.137
	(0.058)	(0.059)	(0.078)
FEMALE	-1.544	-1.743	-1.741
	(0.043)	(0.049)	(0.049)
NONWHITE	-0.155	-0.604	-0.549
	(0.050)	(0.073)	(0.073)
MARRIED	0.905	0.906	0.904
	(0.044)	(0.044)	(0.044)
HIGHSCHOOL	4.613	4.615	4.599
	(0.057)	(0.057)	(0.057)
FEMALE × NONWHITE		0.847	0.851
		(0.100)	(0.100)
REGIONR2			0.216
			(0.073)
REGIONR3			-0.409
			(0.064)
REGIONR4			-0.122
			(0.067)
Num.Obs.	116650	116650	116650
R2	0.067	0.068	0.068
R2 Adj.	0.067	0.067	0.068

F	2446.001	1973.492	1254.706

Comparison of all Models

```
#Interpreting the coefficient estimates for B1, B2, B2 for Column(3)

# In column 3, B1 is -1.741, B2 is -0.549, B3 0.904. This need to be translated to
# Percent change to better understand INCWAGE (not in Log(INCWAGE) term). To do
# this we need to use the Equation ( (e^(B1,B2 or B3) - 1) * 100). We can then see
# that...

# (B1) Females earn about 82.4% less than males.

# (B2) Nonwhite individuals earn about 42.3% less than white individuals.

# (B3) Married individuals earn about 146.7% more than unmarried individuals.
```